Half-Cell Flux and an Exactly Solvable Period-Eight Phase in Signed Cycle Squares

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Abstract

For a fixed graph, changing the edge signs preserves the support of the adjacency matrix while altering its spectrum. We study this optimization problem for the cycle square $C_n(1,2)$. A switching-invariant half-cell flux condition is shown to be equivalent to a natural monomial chiral involution for every even-period Hamilton-gauge signing, reducing a $2m$-dimensional Floquet problem to an m-dimensional squared problem. Its first realization with squared Bloch edge below 8 occurs at primitive period eight. For the resulting antipodal two-defect phase we determine all four squared dispersion branches and prove, for every $L \geq 1$,

$$\rho(A_{8L,+})^2 = 4 + \sqrt{10 + 2\sqrt{5}}.$$

The negative-holonomy finite rings also have an explicit spectral radius. At primitive period eight the displayed phase is the unique sub-eight phase up to the natural symmetries, whereas no shorter primitive period is sub-eight. In particular, for every $L \geq 4$ the positive-holonomy phase has smaller spectral radius than the natural twisted signing on $C_{8L}(1,2)$.

Keywords: signed graph; cycle square; spectral radius; switching; Floquet decomposition; chiral symmetry

MSC 2020: 05C50 (primary); 05C22, 15A18 (secondary)

1 Introduction and main results

1.1 From switching to fixed-graph spectral optimization

A signed graph consists of an underlying graph and a choice of sign on each edge. Its adjacency matrix has the same zero pattern as the ordinary adjacency matrix, but its nonzero entries may be 1 or -1. This modest change separates two kinds of information. Individual edge signs depend on the chosen representative: switching at vertices conjugates the adjacency matrix by a diagonal sign matrix and therefore leaves its spectrum unchanged. Products of signs around cycles, by contrast, survive switching. Balance, switching classes, and cycle signs have consequently been central coordinates in signed graph theory since the work of Harary and Zaslavsky [8, 15].

The spectral question has two distinct forms. One may ask whether a graph admits a signing with eigenvalues controlled by a universal benchmark. This viewpoint is fundamental in the theory of two-lifts: Bilu and Linial connected signings with new eigenvalues of lifts, and the interlacing-polynomial method of Marcus, Spielman, and Srivastava produced Ramanujan signings in the bipartite setting [3, 12]. A finer problem fixes the entire underlying graph and asks for the least spectral radius among all its signings,

$$m(G) = \min_{\sigma:E(G)\to\{\pm 1\}} \rho(A_\sigma). \tag{1}$$

Here the support, order, and degree sequence are held constant; the remaining optimization takes place among switching-invariant cycle patterns. This problem appears naturally in the broader signed spectral programme [2], and exact extremal results are known for several restricted signed graph families [4, 7]. The fixed-graph version is nevertheless rigid: an existence theorem for a good signing does not identify the minimum, and a highly symmetric candidate need not be optimal.

1.2 Why cycle squares and periodic signings

We study (1) for

$$C_n(1,2), \qquad V = \mathbb{Z}/n\mathbb{Z}, \qquad i \sim j \iff i - j \equiv \pm 1, \pm 2 \pmod{n}.$$

This graph is the square of the n-cycle when n is in the range considered below. It occupies a useful middle ground. The unsigned graph is circulant, so its spectrum is read from scalar Fourier symbols [6]; recent signed spectral work on nonbipartite Cayley graphs provides a broader neighboring setting [9]. At the same time, every pair of consecutive step-one edges and the corresponding step-two edge forms a triangle. These overlapping triangles make the cycle data local and strongly coupled: changing one edge alters several nearby cycles.

An arbitrary signing destroys one-site translation invariance. A periodic cycle-sign pattern restores translation by a cell, replacing a scalar Fourier symbol with a small Hermitian matrix-valued symbol. This is the finite graph counterpart of the Bloch decomposition for periodic discrete operators. Gauge, flux, and quasi-momentum play the same distinct roles in unit-gain and periodic magnetic graph theory [13, 10, 11]. In the present finite setting no direct-integral machinery is required: a ring consisting of L cells is an orthogonal direct sum of L finite fibers.

These observations point to the correct variables. After switching the step-one Hamilton cycle to a local positive gauge, let τ_i denote the sign of the triangle based at i, and let $\alpha \in \{\pm 1\}$ retain the global Hamilton-cycle holonomy. The finite boundary condition is quasi-periodic, $x_{i+n} = \alpha x_i$. For local square calculations it is convenient to pass once more to

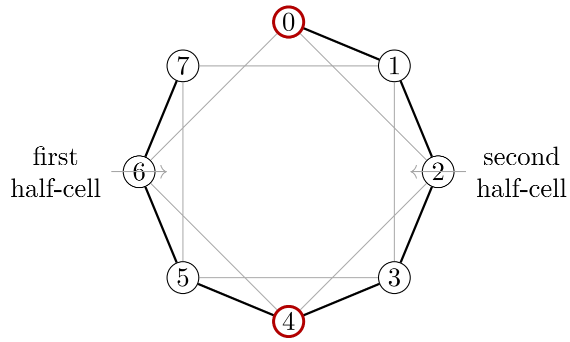
$$Q_i = \tau_i \tau_{i+1}.$$

The word Q records where odd-displacement couplings survive after the operator is squared. Thus periodicity is not imposed to make an enumeration smaller; it is the setting in which local flux, cell translation, and the spectral edge can be compared exactly.

The phase central to this paper has primitive period eight,

$$\tau_* = (1, 1, -1, 1, -1, -1, 1, -1), \qquad Q_* = (1, -1, -1, -1, 1, -1, -1, -1). \tag{2}$$

The two positive entries of Q_* are antipodal. More importantly, the second half of τ_* is the negative of the first. That half-period relation is the source of an anticommuting symmetry, not an accidental property of an 8×8 determinant. Figure 1 displays the cell and the two levels of flux data.



$$\tau_* = (1, 1, -1, 1 \mid -1, -1, 1, -1)$$
$$Q_* = (+, -, -, - \mid +, -, -, -)$$

Figure 1: The period-eight cell. Solid cycle edges are the locally positive step-one gauge; curved chords are step-two edges. The triangle signs form τ_*, while the two marked positions are the positive entries of Q_*. The second half-cell carries the negative of the first τ-word.

1.3 Main results

Let $A^*_{8L,+}$ be the finite signed adjacency matrix obtained by repeating (2) over L cells with positive Hamilton holonomy. Set

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}. \tag{3}$$

Our first result gives an exact finite value rather than an asymptotic band edge or a numerical upper bound.

Theorem 1.1 (Exact finite period-eight phase)**.** *For every integer $L \geq 1$,*

$$\rho(A^*_{8L,+})^2 = \eta.$$

If A^{tw}_{8L} denotes the twisted signing defined in section 4.5, then for every $L \geq 4$,

$$\rho(A^*_{8L,+})^2 = \eta < \rho(A^{\mathrm{tw}}_{8L})^2.$$

Consequently $m(C_{8L}(1,2)) < \rho(A^{\mathrm{tw}}_{8L})$ for every $L \geq 4$.

The exact solution comes from a structural statement valid at every even period. It identifies precisely when the most natural signed half-cell translation is a chiral symmetry. The term *chiral* is used here in its linear-algebraic sense: a self-adjoint involution anticommutes with the Hermitian fiber. Related spectral-symmetry mechanisms occur for signed graph Laplacians and chiral Floquet systems [1, 5]; the criterion below concerns the present signed adjacency operator and its local flux word.

Theorem 1.2 (Half-cell flux criterion)**.** *Let $p = 2m$ and let τ be a $2m$-periodic Hamilton-gauge word. Within the monomial class consisting of the alternating diagonal sign followed by half-period translation, a normalized chiral involution exists on every Bloch fiber if and only if*

$$\tau_{i+m} = -\tau_i \qquad (i \in \mathbb{Z}).$$

Equivalently,

$$Q_{i+m} = Q_i \quad (i \in \mathbb{Z}), \qquad \prod_{j=0}^{m-1} Q_j = -1.$$

Whenever these conditions hold, the characteristic polynomial is even and the squared fiber problem reduces from dimension $2m$ *to dimension* m*.*

The general criterion explains why the period-eight fiber reduces, but does not by itself explain why period eight is selected by the spectral threshold 8. The next theorem supplies that missing extremal statement. Period is understood primitively; repeating a shorter cell does not count as a new phase.

Theorem 1.3 (First occurrence and rigidity)**.** *Eight is the smallest primitive period of a legal Hamilton-gauge signing whose squared Bloch edge is strictly below* 8*. At primitive period eight, the word* Q_* *in* (2) *is the unique sub-eight class, up to cyclic translation, reflection, and the two lifts* τ *and* $-\tau$*. The balanced class has squared edge* 8*, and every other legal period-eight class has edge strictly above* 8*.*

The first three phase-averaged even moments also give a period-independent constraint. If d is the number of positive Q-sites and a, b count positive pairs at cyclic distances one and two, respectively, then an edge at most 8 forces

$$d \leq \frac{3p}{4}, \qquad 40d + 96a + 48b \leq 42p. \tag{4}$$

This does not classify longer words; it identifies the sparse, weakly clustered defect geometry behind the short-period result.

1.4 Position and proof architecture

Theorem 1.1 also resolves a specific recent proposal. The twisted signing was put forward as an optimal candidate for this family in [14]. Our formulation does not rely on that proposal: the fixed-graph optimization problem, the period-eight phase, and the chiral criterion are defined independently. The recent conjecture is one consequence of the strict comparison, rather than the source of the problem.

The proof follows the mathematical dependencies. Section 2 passes from edge signs to (τ, α), proves the symmetry conventions, and derives the finite Bloch sum. Section 3 proves Theorem 1.2 by a coefficient calculation and a flux telescoping identity. Section 4 specializes the reduction, solves the period-eight polynomial, quantizes both finite holonomy sectors, and proves Theorem 1.1. Section 5 asks whether the appearance of period eight is accidental; moment inequalities and exact finite certificates prove Theorem 1.3. Section 6 extracts the general defect obstruction (4), and Section 7 closes with the remaining exact minimization problem.

2 Switching coordinates and periodic fibers

The optimization in (1) is stated in terms of edge signs, but edge signs are not the invariant data. We first pass to local cycle signs and retain the one global sign that cannot be removed on a finite Hamilton cycle. Periodicity of the resulting word then supplies the finite Bloch decomposition used throughout the paper.

2.1 Signed cycle squares and Hamilton gauge

Write s_i for the sign of the step-one edge $\{i, i+1\}$ and t_i for the sign of the step-two edge $\{i, i+2\}$, with indices modulo n. A switching function $\varepsilon : \mathbb{Z}/n\mathbb{Z} \to \{\pm 1\}$ sends

$$s_i \longmapsto \varepsilon_i s_i \varepsilon_{i+1}, \qquad t_i \longmapsto \varepsilon_i t_i \varepsilon_{i+2}.$$

If $D_\varepsilon = \operatorname{diag}(\varepsilon_i)$, the signed adjacency matrices are related by $A \mapsto D_\varepsilon A D_\varepsilon$. Thus switching is an orthogonal similarity.

The sign of the triangle $(i, i+1, i+2)$ is

$$\tau_i = s_i s_{i+1} t_i, \tag{5}$$

and the sign of the step-one Hamilton cycle is

$$\alpha = \prod_{i=0}^{n-1} s_i. \tag{6}$$

Both are switching invariant. Conversely, they determine the switching class. Indeed, choose $\varepsilon_0 = 1$ and then recursively choose $\varepsilon_{i+1} = \varepsilon_i s_i$ along the cut-open Hamilton path. All local step-one signs become positive, the step-two sign based at i becomes τ_i, and the mismatch after one circuit is α.

This construction is best stated on the lift. A vector obeys

$$x_{i+n} = \alpha x_i, \tag{7}$$

and the Hamilton-gauge operator is

$$(A_\tau x)_i = x_{i-1} + x_{i+1} + \tau_{i-2} x_{i-2} + \tau_i x_{i+2}. \tag{8}$$

Thus "positive step-one gauge" is a local normalization on the cut-open lift; negative holonomy is carried by the boundary condition, not by a claim that the product of positive finite-cycle edges is negative.

2.2 Symmetries and primitive period

The local-square word

$$Q_i = \tau_i \tau_{i+1} \tag{9}$$

is invariant under replacing τ by $-\tau$. A p-periodic Q-word has a p-periodic lift precisely when $\prod_{i=0}^{p-1} Q_i = 1$; after τ_0 is chosen, the recurrence $\tau_{i+1} = Q_i \tau_i$ gives the two lifts τ and $-\tau$.

For later orbit reductions we record the spectral effect of this choice. Let $\mathcal{B}_z^{(p)}$ be the space of sequences with $x_{i+p} = z x_i$, where $|z| = 1$, and let $H_\tau^{(p)}(z)$ be the restriction of (8). Set

$$R_p(\tau) = \max_{|z|=1} \rho(H_\tau^{(p)}(z))^2. \tag{10}$$

Proposition 2.1 (Lift and dihedral invariance)**.** *The quantity R_p has the following properties.*

1. $R_p(-\tau) = R_p(\tau)$.
2. *A cyclic translate of τ has the same fiber spectrum at each phase z.*
3. *If $\tau_i^\vee = \tau_{-i-2}$, then* $\operatorname{Spec} H_{\tau^\vee}^{(p)}(z^{-1}) = \operatorname{Spec} H_\tau^{(p)}(z)$.

Consequently R_p is constant on the dihedral orbits of legal Q-words and is independent of the choice of lift.

Proof. Let $(Dx)_i = (-1)^i x_i$. Displacement-one terms change sign under conjugation by D, while displacement-two terms do not, and hence

$$DA_\tau D = -A_{-\tau}.$$

The map D sends $\mathcal{B}_z^{(p)}$ to $\mathcal{B}_{(-1)^p z}^{(p)}$. Negating an operator leaves squared eigenvalues unchanged, and $z \mapsto (-1)^p z$ permutes the unit circle. This proves the first assertion.

For the other two, use $(S_k x)_i = x_{i+k}$ and $(\mathcal{R}x)_i = x_{-i}$. Substitution in (8) gives

$$A_{\mathrm{rot}_k \tau} S_k = S_k A_\tau, \qquad A_{\tau^\vee}\mathcal{R} = \mathcal{R} A_\tau.$$

The shift S_k preserves $\mathcal{B}_z^{(p)}$, whereas $\mathcal{R}$ maps it to $\mathcal{B}_{z^{-1}}^{(p)}$. Both maps are unitary. The induced actions on Q may change the chosen lift only by a global minus sign, which has already been handled. □

A displayed cell need not be primitive. If τ has least period q but is displayed in a cell of length $p = kq$, translation by q further diagonalizes the p-fiber and gives the zone-folding identity

$$\operatorname{Spec} H_\tau^{(p)}(z) = \bigcup_{w^k = z} \operatorname{Spec} H_\tau^{(q)}(w), \tag{11}$$

with multiplicities. Taking the union over $|z| = 1$ proves that cell repetition does not change R_p. We therefore use *primitive period* for the least period of τ and reserve *displayed period* for the chosen cell size.

2.3 Finite Bloch decomposition

Suppose now that τ has displayed period p and $n = pL$. Group the coordinates into L cells,

$$X_r = (x_{rp}, x_{rp+1}, \ldots, x_{rp+p-1})^\mathsf{T}, \qquad 0 \le r < L.$$

The cell shift is unitary and its L eigenvalues are precisely the phases satisfying $z^L = \alpha$. Its eigenspace with eigenvalue z consists of $X_r = z^r X_0$. Since the Hamilton-gauge operator commutes with the cell shift, these eigenspaces are invariant and its restriction is the $p \times p$ Hermitian matrix $H_\tau^{(p)}(z)$.

Proposition 2.2 (Finite Bloch sum)**.** *For every $L \ge 1$ and $\alpha \in \{\pm 1\}$,*

$$A_{pL,\alpha}(\tau) \simeq \bigoplus_{z^L = \alpha} H_\tau^{(p)}(z). \tag{12}$$

In particular,

$$\rho(A_{pL,\alpha}(\tau)) = \max_{z^L = \alpha} \rho(H_\tau^{(p)}(z)). \tag{13}$$

Proof. The eigenspaces of the unitary cell shift are mutually orthogonal, each has dimension p, and there are L of them. Their dimensions sum to pL, so the restrictions give the full matrix with no missing multiplicities. For $|z| = 1$, every boundary-crossing entry is paired with its complex conjugate; hence each fiber is Hermitian. The spectral-radius identity follows from the orthogonal direct sum. □

Figure 2 summarizes the three roles that must remain separate: τ specifies the repeated local operator, α specifies the finite seam, and z labels a cell-shift eigenspace.

The decomposition reduces a periodic signing to a family of finite Hermitian matrices. The next question is structural: which flux patterns force these fiber spectra to be symmetric about zero? The half-cell criterion answering this question is independent of period eight.

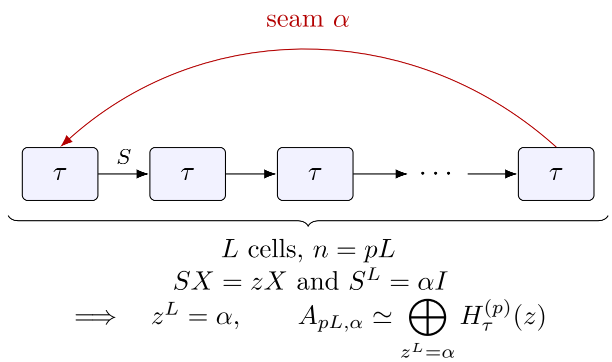


Figure 2: A finite ring of L periodic cells. Local coefficients repeat from cell to cell, the seam carries the Hamilton holonomy α, and diagonalizing the cell shift gives the allowed phases $z^L = \alpha$.

3 Half-cell chiral symmetry

The finite Bloch sum reduces dimension but does not normally simplify an individual fiber. A further reduction becomes possible when half-period translation reverses the step-two coefficient word. The alternating sign needed to reverse the step-one part then produces an anticommuting monomial operator. We now characterize this mechanism within that natural class.

3.1 The half-cell operator

Let $p = 2m$. On the lifted sequence space define

$$(Dx)_i = (-1)^i x_i, \qquad (T_m x)_i = x_{i+m}, \qquad K_m = DT_m. \tag{14}$$

The operator D reverses every displacement-one coupling and preserves every displacement-two coupling. Translation by m compares the two half-cells. Their product is therefore the first monomial candidate for an operator that anticommutes with A_τ.

Theorem 3.1 (Coefficient criterion)**.** *For a $2m$-periodic sign word τ, the following are equivalent:*

1. *$K_m A_\tau = -A_\tau K_m$;*
2. *$\tau_{i+m} = -\tau_i$ for every $i \in \mathbb{Z}$.*

Proof. Apply the anticommutator to an arbitrary sequence. The displacement-one terms cancel, and direct collection of the two remaining coordinates gives

$$\begin{aligned}((A_\tau K_m + K_m A_\tau)x)_i = (-1)^i \big[&(\tau_{i-2} + \tau_{i+m-2})x_{i+m-2} \\ &+ (\tau_i + \tau_{i+m})x_{i+m+2}\big].\end{aligned} \tag{15}$$

If the second half-word is the negative of the first, both coefficients vanish. Conversely, if the anticommutator vanishes on every sequence, the coefficient of x_{i+m+2} vanishes for every i, which gives $\tau_{i+m} = -\tau_i$. The other coefficient then vanishes as well. □

The operator K_m preserves every Bloch space $\mathcal{B}_z^{(2m)}$, but it is not yet an involution. On this space,

$$T_m^2 = zI, \qquad T_m D = (-1)^m DT_m, \qquad K_m^2 = (-1)^m zI. \tag{16}$$

Choose a unit scalar $\gamma_m(z)$ satisfying

$$\gamma_m(z)^2 = (-1)^m z^{-1}. \tag{17}$$

If $\xi^2 = z$, one may take $\gamma_m(z) = \xi^{-1}$ for even m and $\gamma_m(z) = \mathrm{i}\xi^{-1}$ for odd m. Then

$$J_z = \gamma_m(z) K_m|_{\mathcal{B}_z^{(2m)}} \tag{18}$$

is unitary and satisfies $J_z^2 = I$; hence $J_z = J_z^*$. Changing the square root of z replaces J_z by $-J_z$ and only interchanges its two eigenspaces.

3.2 The flux criterion

The coefficient condition is transparent in τ, but Q is the local word used in the extremal analysis. The next equivalence shows that chirality is visible without choosing either lift of Q.

Theorem 3.2 (Negative half-cell flux)**.** *For a $2m$-periodic sign word τ, the following are equivalent:*

1. *$\tau_{i+m} = -\tau_i$ for every i;*
2. *$Q_{i+m} = Q_i$ for every i and $\prod_{j=0}^{m-1} Q_j = -1$.*

Under these conditions the operators J_z in (18) are chiral involutions:

$$J_z H_\tau^{(2m)}(z) = -H_\tau^{(2m)}(z) J_z \qquad (|z| = 1). \tag{19}$$

Proof. If $\tau_{i+m} = -\tau_i$, then

$$Q_{i+m} = (-\tau_i)(-\tau_{i+1}) = Q_i, \qquad \prod_{j=0}^{m-1} Q_j = \tau_0\tau_m = -1.$$

For the converse, put $r_i = \tau_{i+m}/\tau_i \in \{\pm 1\}$. Half-periodicity of Q gives $r_i r_{i+1} = 1$, so all r_i equal one constant r. Telescoping over a half-cell gives

$$r = \frac{\tau_m}{\tau_0} = \prod_{j=0}^{m-1} Q_j = -1.$$

Thus $\tau_{i+m} = -\tau_i$. The anticommutation follows from Theorem 3.1; scalar normalization does not change it. ☐

Figure 3 records the mechanism: the alternating diagonal sign reverses the nearest-neighbor part, while negative half-cell flux forces translation to reverse the step-two coefficients.

3.3 Algebraic consequences

The monomial matrix of J_z has no fixed coordinate, so its trace is zero. As an involution on a $2m$-dimensional space, its $+1$ and -1 eigenspaces therefore both have dimension m. Equation (19) sends either eigenspace to the other. In a unitary basis adapted to this decomposition,

$$H_\tau^{(2m)}(z) \simeq \begin{pmatrix} 0 & B(z)^* \\ B(z) & 0 \end{pmatrix}, \qquad H_\tau^{(2m)}(z)^2 \simeq \begin{pmatrix} B(z)^*B(z) & 0 \\ 0 & B(z)B(z)^* \end{pmatrix}. \tag{20}$$

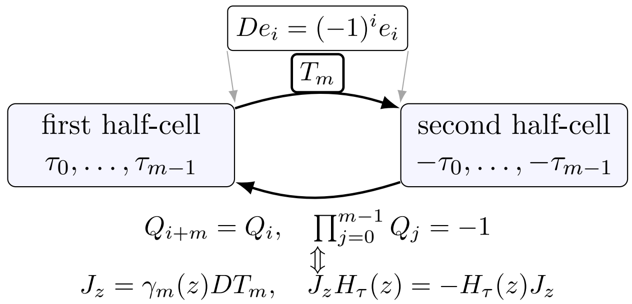


Figure 3: The half-cell chiral mechanism. Translation by m exchanges the two half-cells; the alternating diagonal factor reverses step-one couplings. The local condition $\tau_{i+m} = -\tau_i$, equivalently half-periodic Q with negative half-cell flux, reverses the step-two couplings as well.

Consequently the spectrum is symmetric about zero and

$$\det(\lambda I_{2m} - H_\tau^{(2m)}(z)) = \det(\lambda^2 I_m - B(z)^* B(z)). \tag{21}$$

The theorem does not make the general $m \times m$ block explicitly solvable. Its role is to locate a structural class in which half of the spectral dimension disappears. The phase (2) belongs to this class with $m = 4$. It is also the first member of the class known here to cross the squared edge 8. We now solve that fiber completely.

4 The exact period-eight phase

Theorem 3.2 explains a dimension-halving mechanism but does not solve the resulting block at a general period. The word τ_* is the first sub-eight phase and has enough additional structure to reduce its 4×4 squared block once more. This section carries that calculation through to the complete dispersion and then returns to the finite rings.

4.1 The target fiber and its chiral block

For $|z| = 1$, the period-eight fiber in the residue order $0, 1, \dots, 7$ is

$$H(z) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & z^{-1} & z^{-1} \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & -z^{-1} \\ 1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & -1 \\ z & 0 & 0 & 0 & -1 & 1 & 0 & 1 \\ z & -z & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}. \tag{22}$$

The boundary entries occur in conjugate pairs, so $H(z)$ is Hermitian. Since $\tau_{i+4} = -\tau_i$, Section 3 applies. Choose ξ with $\xi^2 = z$. Here $m = 4$, so $\gamma_4(z) = \xi^{-1}$, and the normalized signed four-site shift is the chiral involution J_z.

In bases of its two eigenspaces, $H(z)$ is off diagonal. Multiplying the two off-diagonal blocks gives the following squared block:

$$S(\xi) = \begin{pmatrix} 4-s & 0 & 1+\xi^{-1} & 2 \\ 0 & 4-s & 2 & 1-\xi^{-1} \\ 1+\xi & 2 & 4+s & 0 \\ 2 & 1-\xi & 0 & 4+s \end{pmatrix}, \qquad s = \xi + \xi^{-1}. \tag{23}$$

The squared eigenvalues of $H(z)$ are the eigenvalues of $S(\xi)$, each occurring in both chiral blocks. Replacing ξ by $-\xi$ exchanges the two chiral coordinate spaces; the final characteristic data depend only on z.

4.2 The period-eight polynomial

Write (23) in 2×2 blocks as

$$S(\xi) = \begin{pmatrix} (4-s)I_2 & R(\xi) \\ Q(\xi) & (4+s)I_2 \end{pmatrix},$$

where

$$R(\xi) = \begin{pmatrix} 1+\xi^{-1} & 2 \\ 2 & 1-\xi^{-1} \end{pmatrix}, \qquad Q(\xi) = \begin{pmatrix} 1+\xi & 2 \\ 2 & 1-\xi \end{pmatrix}.$$

Because the diagonal blocks are scalar, the elementary block determinant identity gives, without an exceptional Schur-complement value,

$$\det(yI_4 - S(\xi)) = \det\Big(((y-4)^2 - s^2)I_2 - R(\xi)Q(\xi)\Big). \tag{24}$$

Using $\xi^2 = z$ and $c = z + z^{-1} = \xi^2 + \xi^{-2}$, expansion of the 2×2 determinant yields

$$\begin{aligned} P(y,c) = y^4 &- 16y^3 + (80-2c)y^2 \\ &+ (-128+16c)y + c^2 - 13c + 38. \end{aligned} \tag{25}$$

Thus

$$\det(\lambda I_8 - H(z)) = P(\lambda^2, z + z^{-1}). \tag{26}$$

This identity is the full bridge from the general chiral theorem to an exact period-eight spectrum.

4.3 Complete dispersion

Put $y = X + 4$. The quartic becomes even in X:

$$P(X+4, c) = X^4 - (16+2c)X^2 + c^2 + 19c + 38. \tag{27}$$

With $W = X^2$, the two roots are

$$W_\tau(c) = 8 + c + \tau\sqrt{26-3c}, \qquad \tau \in \{\pm 1\}. \tag{28}$$

For $-2 \le c \le 2$, both are positive. Indeed,

$$(8+c)^2 - (26-3c) = c^2 + 19c + 38 \ge 4,$$

and $8 + c > 0$. The four squared branches are therefore

$$y_{\sigma,\tau}(c) = 4 + \sigma\sqrt{W_\tau(c)}, \qquad \sigma, \tau \in \{\pm 1\}. \tag{29}$$

Since $0 < W_- < W_+ < 16$, their order is uniform:

$$y_{+,+} > y_{+,-} > y_{-,-} > y_{-,+} > 0.$$

Moreover,

$$W_+'(c) = 1 - \frac{3}{2\sqrt{26-3c}} > 0, \qquad W_-'(c) = 1 + \frac{3}{2\sqrt{26-3c}} > 0.$$

Thus the two upper branches increase with c, while the two lower branches decrease. In particular the top squared edge is

$$r(c) = 4 + \sqrt{8 + c + \sqrt{26-3c}}, \tag{30}$$

strictly increasing on $[-2, 2]$.

For completeness, set

$$a = \sqrt{10 + 2\sqrt{5}}, \qquad b = \sqrt{10 - 2\sqrt{5}}.$$

At $c = -2$ the four squared roots are

$$6 + \sqrt{2}, \quad 6 - \sqrt{2}, \quad 2 + \sqrt{2}, \quad 2 - \sqrt{2},$$

whereas at $c = 2$ they are $4 + a, 4 + b, 4 - b, 4 - a$. The four squared bands are therefore

$$\begin{array}{ll} [6+\sqrt{2},\, 4+a], & [6-\sqrt{2},\, 4+b], \\ [4-b,\, 2+\sqrt{2}], & [4-a,\, 2-\sqrt{2}]. \end{array} \tag{31}$$

They are disjoint. Equation (26) then recovers the eight simple Hermitian eigenvalues

$$\{\, \pm\sqrt{y_{\sigma,\tau}(c)} : \sigma, \tau \in \{\pm 1\}\,\}.$$

In particular, the spectrum has the exact central gap

$$\left(-\sqrt{4-a}, \sqrt{4-a}\right). \tag{32}$$

4.4 Finite phase quantization

The infinite unit-circle maximum occurs uniquely at $c = 2$, or $z = 1$, and is

$$r(2) = 4 + a = \eta.$$

For positive holonomy the finite grid $z^L = 1$ always contains $z = 1$. Proposition 2.2 therefore gives both an upper bound and an attaining fiber:

$$\rho(A^*_{8L,+})^2 = \eta \qquad (L \geq 1). \tag{33}$$

The two edge eigenvalues $\pm\sqrt{\eta}$ are simple in $H(1)$ because the four squared roots at $c = 2$ are distinct.

For negative holonomy, the allowed phases are

$$z_j = \exp\left(\frac{(2j+1)\pi\mathrm{i}}{L}\right).$$

The largest value of $z + z^{-1}$ is $2\cos(\pi/L)$. Strict monotonicity of r gives the second exact finite formula

$$\rho(A^*_{8L,-})^2 = 4 + \sqrt{8 + 2\cos(\pi/L) + \sqrt{26 - 6\cos(\pi/L)}}. \tag{34}$$

It is strictly below η for every finite L and increases to η as $L \to \infty$. Both holonomy sectors are therefore samples of one dispersion law; their difference is entirely the finite phase grid.

4.5 The twisted comparison

We now compare the exactly solved phase with the natural twisted signing. In Hamilton-gauge coordinates, this is the alternating triangle-flux class with negative Hamilton holonomy. Its two-site Fourier block is

$$B(t) = 2\begin{pmatrix} \cos t & \cos 2t \\ \cos 2t & -\cos t \end{pmatrix}, \tag{35}$$

so its eigenvalues have squared magnitude $4g(t)$, where $g(t) = \cos^2 t + \cos^2 2t$. The shifted finite grid is $t = (2k+1)\pi/(8L)$. By $g(-t) = g(t) = g(\pi - t)$ it is enough to consider $t \in [\pi/(8L), \pi/2]$. With $u = \cos^2 t$,

$$g(t) = 4u^2 - 3u + 1.$$

Thus $g(t) \le 1$ on $[\pi/6, \pi/2]$. On $(0, \pi/6]$,

$$g'(t) = -\sin(2t)(1 + 4\cos(2t)) < 0.$$

Finally $g(\pi/(8L)) \ge g(\pi/8) = (4+\sqrt{2})/4 > 1$. Hence the first positive grid phase is maximal, and

$$\rho(A^{\mathrm{tw}}_{8L})^2 = 4 + 2\cos\frac{\pi}{4L} + 2\cos\frac{\pi}{2L}. \tag{36}$$

Proof of Theorem 1.1. It remains to compare (33) and (36). The latter is increasing in $L \ge 4$, so it is enough to use $L = 4$. We record an exact rational separator only for this short comparison:

$$\eta < \frac{1561}{200} < 4 + 2\cos\frac{\pi}{16} + 2\cos\frac{\pi}{8}. \tag{37}$$

For the first inequality, $\sqrt{5} < 2239/1000$ and

$$10 + 2\sqrt{5} < \frac{7239}{500} < \left(\frac{761}{200}\right)^2.$$

For the second, repeated half-angle identities give

$$2\cos\frac{\pi}{16} = \sqrt{2+\sqrt{2+\sqrt{2}}}, \qquad 2\cos\frac{\pi}{8} = \sqrt{2+\sqrt{2}}.$$

The elementary square comparisons

$$\sqrt{2+\sqrt{2}} > \frac{923}{500}, \qquad \sqrt{2+\sqrt{2+\sqrt{2}}} > \frac{1961}{1000}$$

make the right-hand side of (37) larger than $4 + 3807/1000 > 1561/200$. This proves the strict comparison for every $L \ge 4$. The conclusion for $m(C_{8L}(1,2))$ follows by using $A^*_{8L,+}$ as an admissible signing. □

The rational number in (37) is not a spectral constant; it only separates two exact values. A supplementary Lean development checks the $\alpha = +1$ finite comparison through this separator in Hermitian eigenvalue form. The complete dispersion and (34) are analytic results independently checked by exact symbolic scripts, not claims about the scope of that formalization.

The exact solution establishes a low-edge phase. It does not yet explain why the first such phase has period eight or whether several unrelated period-eight words share the same behavior. Those are the two questions of the next section.

5 First occurrence and rigidity

The preceding section gives a distinguished period-eight phase, but an exact solution of one word does not explain the period. We now ask two successive questions. Can a shorter primitive word have squared Bloch edge below 8? If not, how many low-edge phases occur at the first feasible period? A local square identity and three trace moments reduce both questions to a small number of exact configurations.

5.1 Local squares and moment reduction

Squaring (8) makes the role of Q visible. Direct substitution gives

$$\begin{aligned}(A_\tau^2 x)_i = 4x_i + x_{i-2} + x_{i+2} + \tau_{i-4}\tau_{i-2}x_{i-4} + \tau_i\tau_{i+2}x_{i+4} \\ + \tau_{i-3}(1+Q_{i-3})x_{i-3} + \tau_{i-2}(1+Q_{i-2})x_{i-1} \\ + \tau_{i-1}(1+Q_{i-1})x_{i+1} + \tau_i(1+Q_i)x_{i+3}.\end{aligned} \tag{38}$$

Thus $Q_i = -1$ cancels the corresponding odd-displacement coupling, while $Q_i = 1$ activates a term of absolute value two. We call a positive Q-site a *defect*.

For a p-periodic word, define the phase-averaged even moments

$$M_k(\tau) = \frac{1}{2\pi}\int_0^{2\pi} \operatorname{tr}\Big(H_\tau^{(p)}(e^{\mathrm{i}\theta})^{2k}\Big)\, d\theta. \tag{39}$$

Let d be the number of positive entries of Q, and let

$$a = \#\{i : Q_i = Q_{i+1} = 1\}, \qquad b = \#\{i : Q_i = Q_{i+2} = 1\}, \tag{40}$$

with cyclic indices. Taking the diagonal of (38), then the squared row norm, gives the first two moments. For the third, expand $\sum_{i,j,k}(A_\tau^2)_{ij}(A_\tau^2)_{jk}(A_\tau^2)_{ki}$ using the same nine displacements; translation-equivalent terms group according to one positive site, an adjacent positive pair, or a distance-two positive pair. The result is

$$M_1 = 4p, \qquad M_2 = 20p + 16d, \qquad M_3 = 118p + 168d + 96a + 48b. \tag{41}$$

For M_2, $(1+Q_i)^2 = 2(1+Q_i)$ and every Q_i occurs in four odd-displacement row entries, producing $16d$. The coefficients in M_3 are the grouped weights of the three local types just listed.

If $R_p(\tau) \le 8$, every squared fiber eigenvalue y satisfies $0 \le y \le 8$. Hence $y^{k+1} \le 8y^k$, and averaging gives $M_{k+1} \le 8M_k$. Applying this for $k = 1, 2$ yields

$$16d - 12p \le 0, \qquad 40d + 96a + 48b - 42p \le 0. \tag{42}$$

These inequalities perform the structural reduction before any finite case is checked.

Table 1: Exact Rayleigh certificates after the moment and dihedral reductions. Each final column is the exact value of $v^*(8I - H^2)v$.

p	Q	z	v	value
5	$----+$	1	$(1,1,1,0,1)$	-4
6	$----++$	1	$(1,1,1,0,1,1)$	-12
6	$---+-+$	-1	$(0,0,2,-1,2,-2)$	-4
6	$--+--+$	i	$(3+\mathrm{i},3,3+2\mathrm{i},1+3\mathrm{i},2+2\mathrm{i},4\mathrm{i})$	-4
7	$------+$	1	$(1,1,1,0,1,0,1)$	-2
7	$---+-++$	1	$(1,1,1,0,0,1,1)$	-4
7	$--+--++$	1	$(1,1,1,1,1,1,1)$	-8
7	$--+-+-+$	-1	$(1,1,1,1,1,1,-1)$	-8

5.2 Periods below eight

A legal Q-word satisfies $\prod_i Q_i = 1$. Combining legality, (42), and the dihedral invariance of Proposition 2.1 leaves the following representatives for $p < 8$:

p	representatives not excluded by the moments
1	none
2	$--$
3	$--+$
4	$----$
5	$----+$
6	$------$, $----++$, $---+-+$, $--+--+$
7	$------+$, $---+-++$, $--+--++$, $--+-+-+$

(43)

The all-negative words have even period; their lifts alternate and have squared edge exactly 8. Every other row is closed by the next exact lemma.

Choose the lift with $\tau_0 = 1$, and put

$$C_\tau(z) = 8I - H_\tau^{(p)}(z)^2.$$

A vector v with $v^* C_\tau(z) v < 0$ proves that the corresponding squared fiber has an eigenvalue above 8.

Lemma 5.1 (Exact closure of the short-period survivors)**.** *For the $p = 3$ representative in* (43)*, at $z = e^{\pi \mathrm{i}/3}$,*

$$\det C_\tau(z) = -1.$$

For every remaining nonbalanced representative, Table 1 gives a vector v and a phase z for which $v^* C_\tau(z) v < 0$.

Proof. Each row is a direct multiplication in a matrix of order at most seven. The displayed value is negative, so $C_\tau(z)$ is not positive semidefinite and $H_\tau^{(p)}(z)^2$ has an eigenvalue larger than 8. For $p = 3$, the matrix $C_\tau(e^{\pi \mathrm{i}/3})$ is Hermitian and has negative determinant, so it also has a negative eigenvalue. These checks exhaust the representatives in (43); no numerical spectral approximation is used. □

Theorem 5.2 (Smallest primitive period)**.** *No legal word of primitive period below eight has squared Bloch edge below* 8*. The word τ_* has primitive period eight and edge $\eta < 8$.*

Proof. The displayed-period claim follows from (43), Lemma 5.1, and the exact edge of the balanced alternating word. Equation (11) shows that repeating a cell does not change its Bloch edge. Finally, $\tau_{i+4} = -\tau_i$ excludes period four, and its first four signs exclude period one or two; hence τ_* is primitive of period eight. □

5.3 Rigidity at period eight

At the first feasible period, the same reduction is even sharper. Applying legality and (42) with $p = 8$ leaves, up to dihedral symmetry, the balanced word and four two-defect words. One has antipodal defects and is Q_*. The other three have cyclic defect separations $1, 2, 3$.

The balanced lift is the alternating period-two phase and has edge 8. The antipodal word has edge $\eta < 8$ by Section 4. It remains to rule out the three nonantipodal pairs. This can be done by one exact closed-walk recurrence rather than three spectral computations. For a starting residue r, let $f_\ell^{(r)}(j)$ be the signed walk sum of length ℓ from r to j on the periodic lift, with $f_0^{(r)}(j) = \delta_{rj}$. Then

$$f_{\ell+1}^{(r)}(j) = f_\ell^{(r)}(j-1) + f_\ell^{(r)}(j+1) + \tau_{j-2} f_\ell^{(r)}(j-2) + \tau_j f_\ell^{(r)}(j+2). \qquad (44)$$

Let M_k be the phase-averaged moment (39) and set $E_k = M_{k+1} - 8M_k$. If the squared edge were at most 8, then $E_k \le 0$ for every k. For defect separations $1, 2, 3$, respectively, recurrence (44) gives the first positive excesses

$$E_4 = 5504, \qquad E_6 = 64336, \qquad E_9 = 2872096. \qquad (45)$$

All quantities are integers obtained by the displayed recurrence. Thus each nonantipodal word has edge above 8.

Theorem 5.3 (Period-eight trichotomy)**.** *Among legal period-eight Q-words, modulo translation, reflection, and lift, the antipodal two-defect class Q_* is the unique class with squared Bloch edge below* 8*. The balanced class has edge* 8*, and every other class has edge above* 8*.*

Proof. The moment reduction leaves precisely the five classes described above. The balanced and antipodal values are exact, and (45) excludes the other three. Proposition 2.1 justifies the orbit quotient and the choice of lift. □

Theorem 5.2 and Theorem 5.3 together prove Theorem 1.3. Period eight is not merely the first cell size at which one selected example works: the low-edge orbit at that first period is rigid.

The proof suggests a broader question. Is this rigidity only a consequence of checking periods at most eight, or do the local defects of every low-edge periodic word obey common restrictions? The same moments answer the latter question without attempting an all-period classification.

6 Periodic defect obstructions

The moment identities used to reach the finite survivor list did not depend on the period being small. We record their general consequence to separate the local geometry forced by a low edge from the exact classification that is special to period eight.

6.1 Arbitrary periodic words

Let τ have any displayed period p, let $Q_i = \tau_i \tau_{i+1}$, and define d, a, b by (40). The closed-walk calculation behind (41) gives the following statement for every p.

Theorem 6.1 (Moment obstruction)**.** *If $R_p(\tau) \leq 8$, then*

$$d \leq \frac{3p}{4}, \qquad 40d + 96a + 48b \leq 42p. \tag{46}$$

Proof. The phase average selects exactly the closed walks whose net cell translation is zero, so the local expansion gives

$$M_1 = 4p, \qquad M_2 = 20p + 16d, \qquad M_3 = 118p + 168d + 96a + 48b.$$

Under the assumed edge bound, $M_2 \leq 8M_1$ and $M_3 \leq 8M_2$. Substitution and simplification give (46). □

6.2 Density and clustering

The first inequality limits defect density. The second also charges adjacent positive pairs through a and distance-two pairs through b. For example, at a fixed value of d, moving defects together can only make the necessary condition harder to satisfy. This is the period-independent remnant of the cancellation seen after squaring: negative Q-sites suppress odd couplings, whereas positive sites reactivate them, and short-range clustering is detected by the next moment.

These inequalities are necessary, not sufficient. They neither describe all longer-period words with edge at most 8 nor determine the global minimum in (1). Their role is more precise: they explain why the short-period candidate set collapses to sparse defects and why the antipodal period-eight geometry is compatible with a lower edge. No higher-moment catalogue is needed for the results of this paper.

7 Concluding remarks

The signed spectral problem on $C_n(1,2)$ is governed more naturally by cycle flux than by individual edge signs. Negative half-cell flux yields a general monomial chiral involution and halves the squared Bloch problem. Period eight is the first primitive period at which this mechanism produces squared edge below 8, and the resulting antipodal two-defect orbit is unique at that period. Its additional block structure makes the entire dispersion explicit, so both finite holonomy sectors have exact spectral radii. The positive sector then gives an infinite family that strictly improves the twisted benchmark.

The exact value $\sqrt{\eta}$ is an attained radius for the displayed signing, not yet a determination of $m(C_{8L}(1,2))$. The remaining fixed-graph question is therefore sharp: determine whether

$$m(C_{8L}(1,2)) = \sqrt{\eta}$$

for an infinite subfamily of L, and, whenever equality holds, characterize all minimizing switching classes. The half-cell criterion and the defect obstructions provide two structural coordinates for that problem, while the present argument requires no claim about all even orders or all periodic phases.

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